LEARNER GUIDE

Algebraic proof

Algebraic Expressions and Identities · Lesson 7. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

Testing examples gives confidence, but there are infinitely many numbers. Algebra covers them all at once: if n is any whole number, $2n$ is any even number, $2n + 1$ is any odd number, and n, $n + 1$, $n + 2$ are any three consecutive numbers. Work with these, then show the result has the form you need — for example $3(n + 1)$, which is a multiple of 3. That is a **proof**. One **counterexample** is enough to show a claim is false.

What you are learning to notice

- Examples are evidence, not proof: $n^2 + n + 41$ is prime for $n = 0$ to 39, and not for $n = 40$.
- Factorising often finishes a proof: $3n + 3 = 3(n + 1)$ shows a multiple of 3.
- 2(something) + 1 shows an odd number; 2(something) shows an even number.
- Two different numbers need two different letters: $2a + 1$ and $2b + 1$.
- Every proof ends with a sentence saying what has been shown.

Moving through it, one step at a time

Test the claim with a few numbers. If any test fails, that is a counterexample — the claim is false. If every test holds, write the numbers in algebra, expand and simplify, and rearrange into the form you need. Finish with a sentence that says what you have proved.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Test, decide, prove

Claim: the sum of any three consecutive whole numbers is a multiple of 3.

1. Test: 4 + 5 + 6 = 15, 10 + 11 + 12 = 33.	both multiples of 3
2. Write the numbers in algebra.	n, $n + 1$, $n + 2$
3. Add and simplify.	$n + (n + 1) + (n + 2)$ $= 3n + 3$
4. Factorise.	$= 3(n + 1)$
5. Conclude.	3 × a whole number, so always a multiple of 3

Proved: the sum of any three consecutive whole numbers is a multiple of 3. ■

Example 2 — A counterexample

Claim: $n^2 + n + 41$ is always prime.

1. Test $n = 0, 1, 2, 3$: 41, 43, 47, 53.	all prime
2. It stays prime all the way to $n = 39$.	forty examples
3. Test $n = 40$: 1600 + 40 + 41 = 1681.	= 41 × 41
4. 1681 is not prime, so the claim is false.	one counterexample

Not always prime — however many examples work, one counterexample is enough.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (odd and even numbers, multiples, and expanding and factorising), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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